PRACTICAL 9 : ENTERPRISE NETWORK DEPLOYMENT AND TROUBLESHOOTING

WEEK 9

Objective

Deploy and troubleshoot a complete Cisco-based enterprise network.

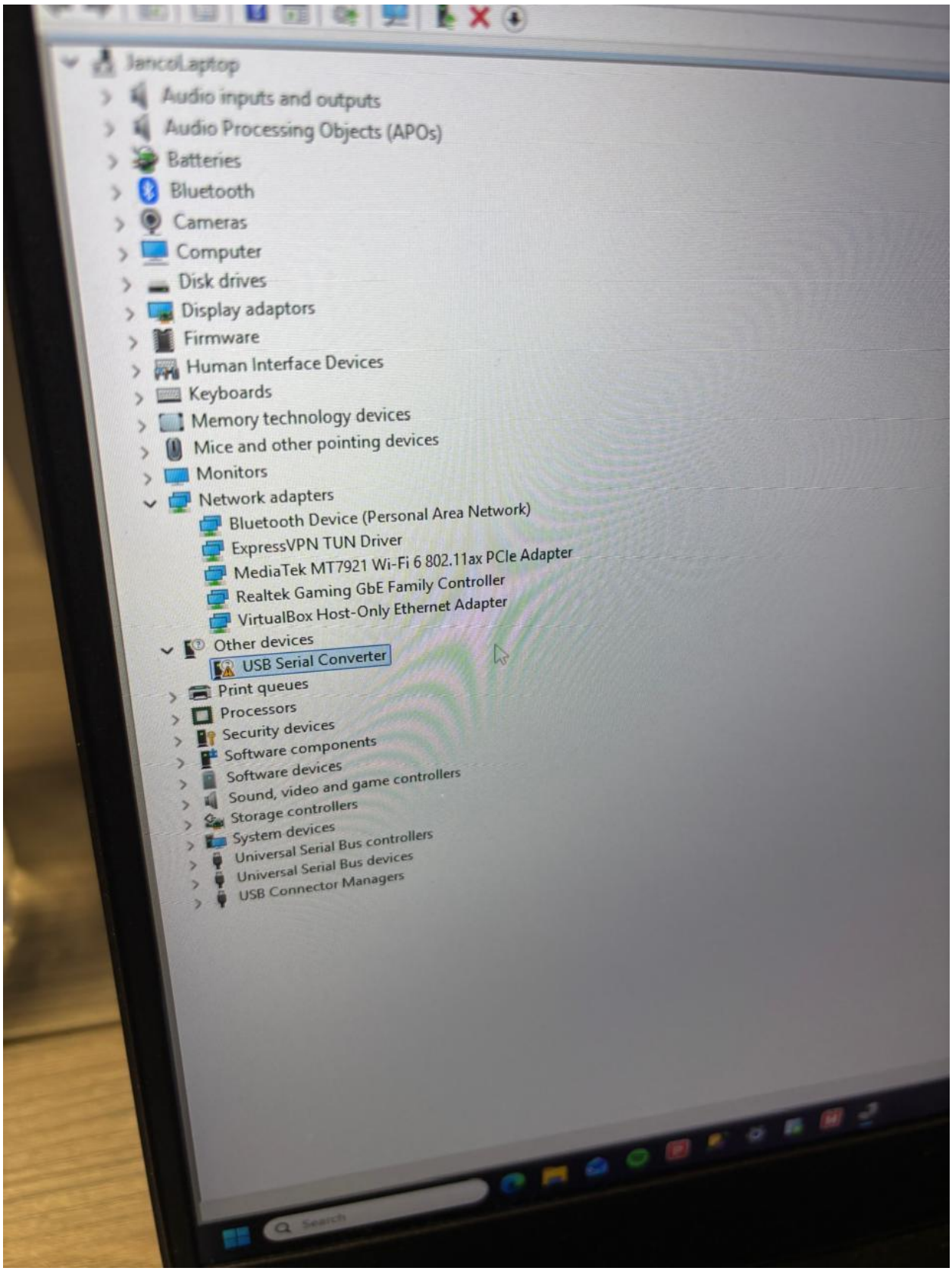
Equipment

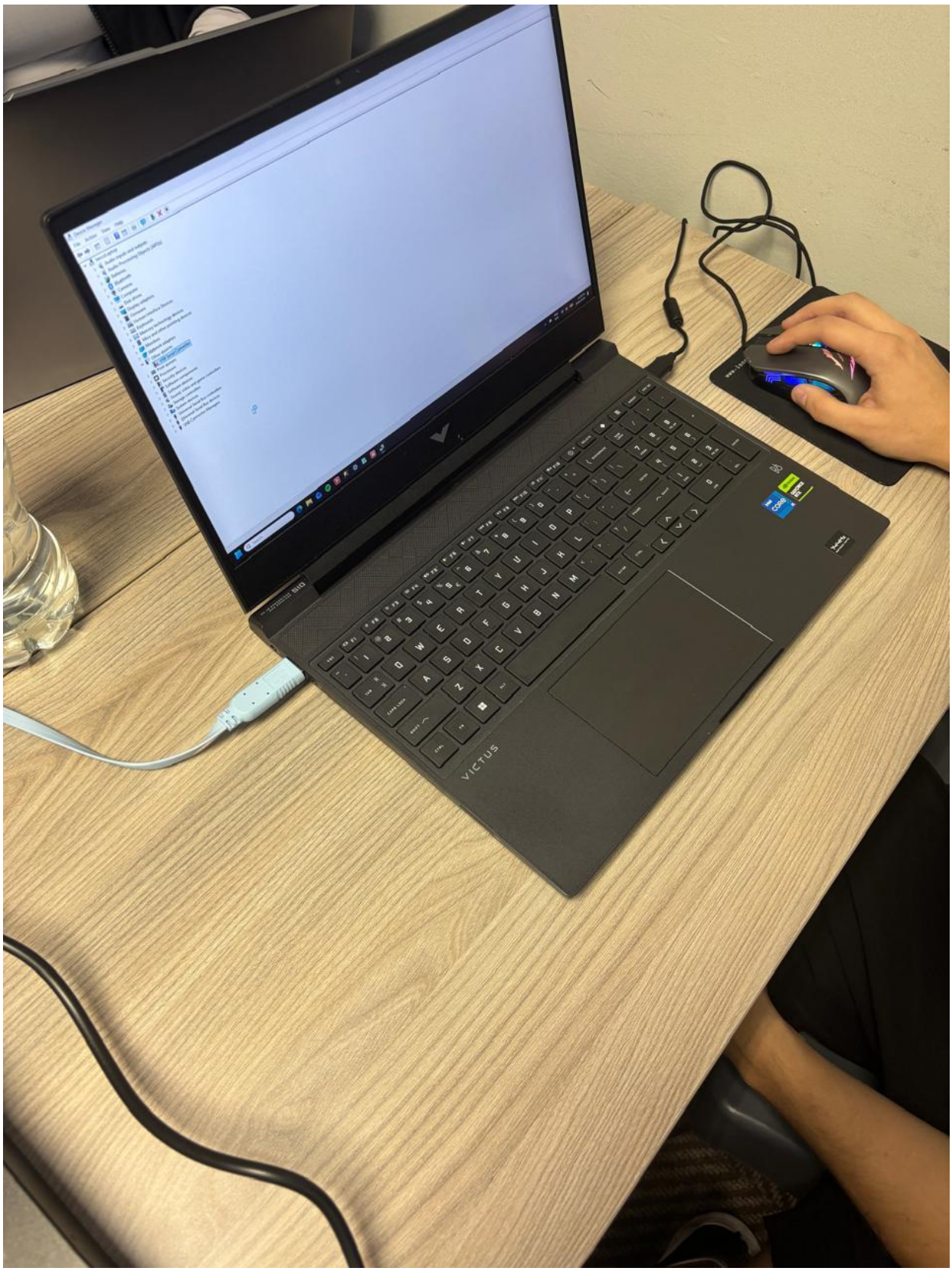
- Cisco Routers
- Cisco Switches
- Switch Cabinets
- Patch Panels

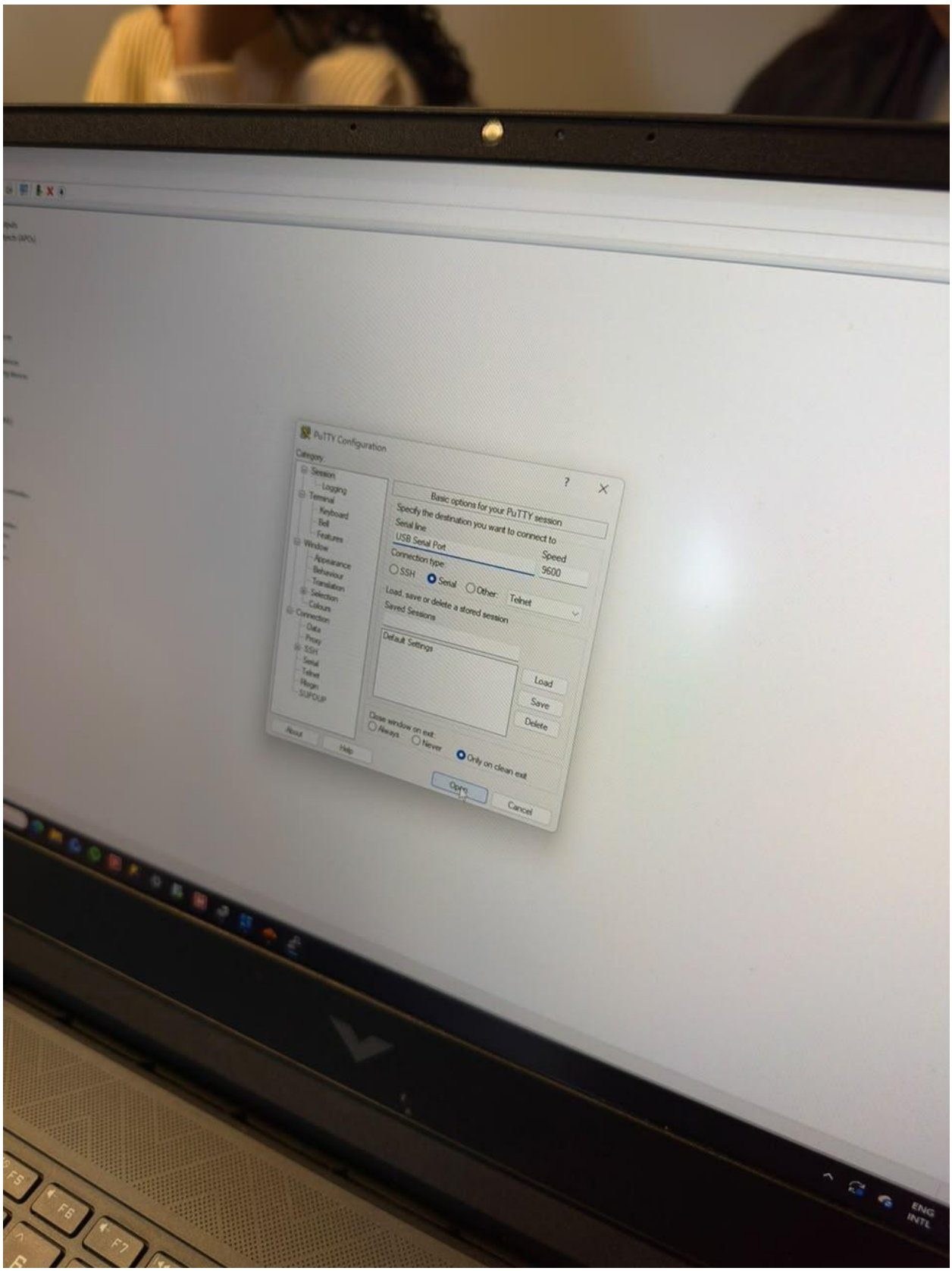
- Printers
- 10 PCs

Activities

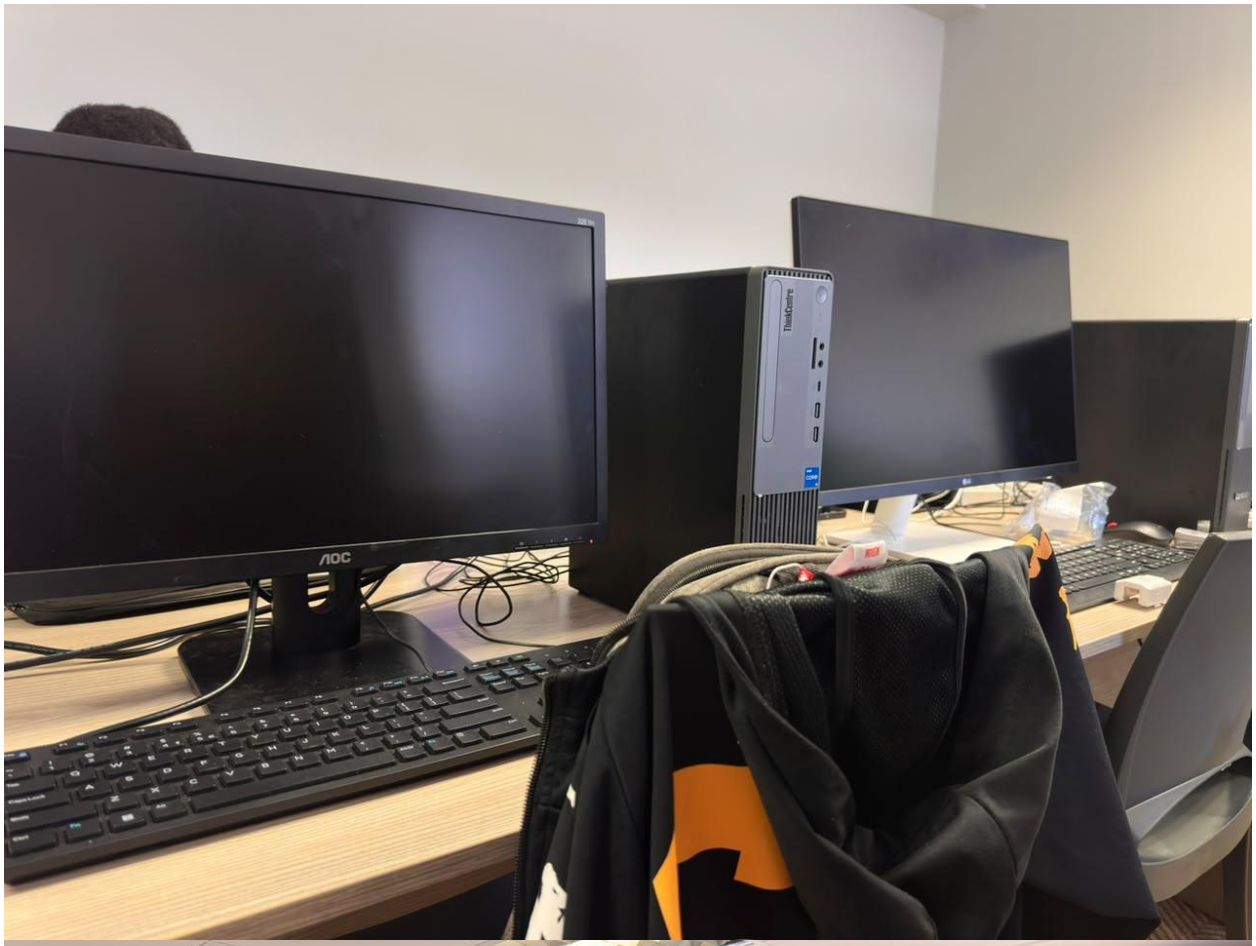



JancoLaptop
Audio inputs and outputs
Audio Processing Objects (APOs)
Batteries
Bluetooth
Cameras
Computer
Disk drives
Display adaptors
Firmware
Human Interface Devices
Keyboards
Memory technology devices
Mice and other pointing devices
Monitors
Network adapters
Bluetooth Device (Personal Area Network)
ExpressVPN TUN Driver
MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
Realtek Gaming GbE Family Controller
VirtualBox Host-Only Ethernet Adapter
Other devices
USB Serial Converter
Print queues
Processors
Security devices
Software components
Software devices
Sound, video and game controllers
Storage controllers
System devices
Universal Serial Bus controllers
Universal Serial Bus devices
USB Connector Managers

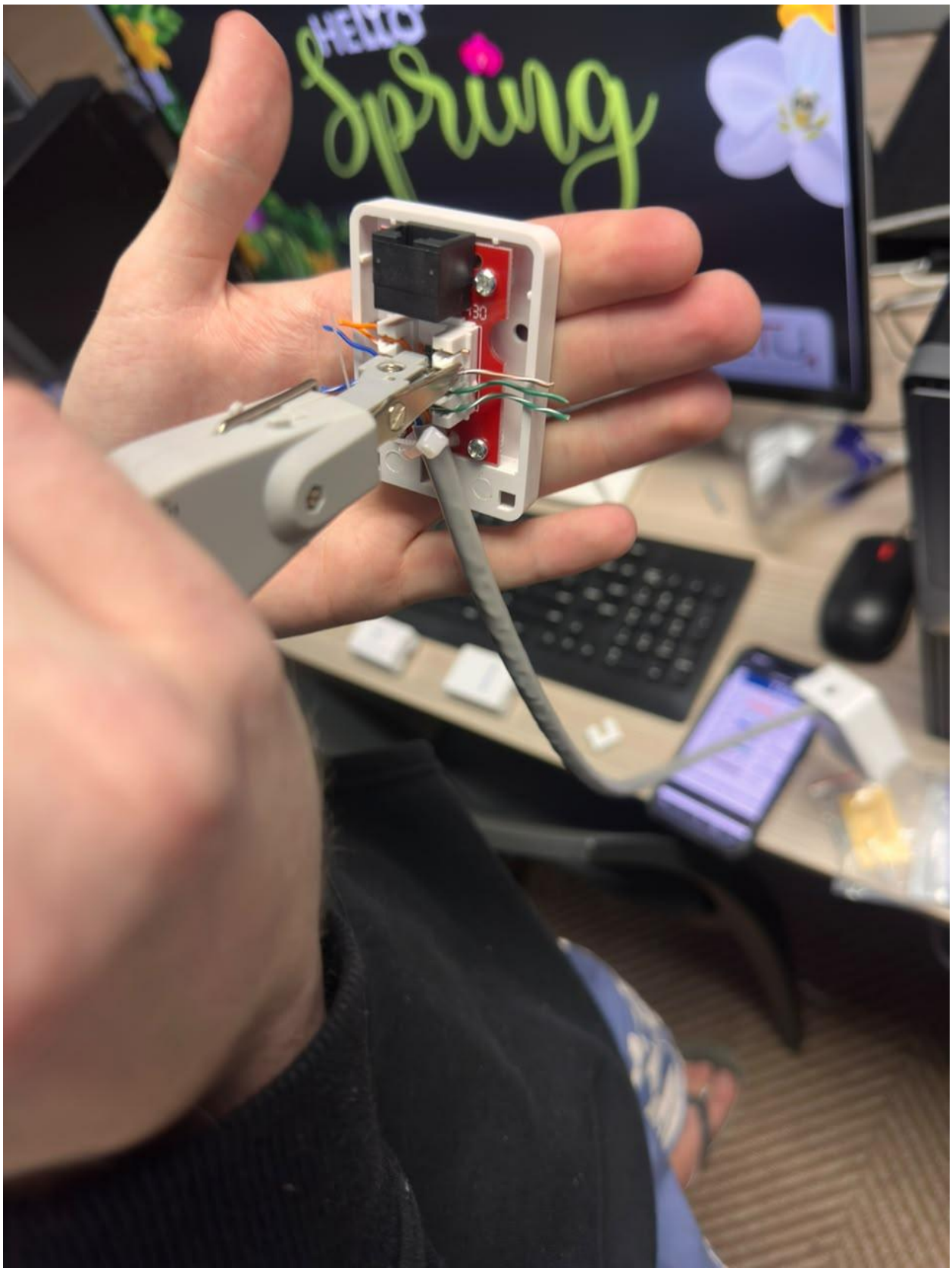

victus


PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Serial
Telnet
Rlogin
SUPDUP
Basic options for your PuTTY session
Specify the destination you want to connect to
Serial line
Speed
USB Serial Port
9600
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Default Settings
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel
ENG
INTL


CTU
training solutions
10:57 AM
10/2/2026


AOC




HELLO
Spring



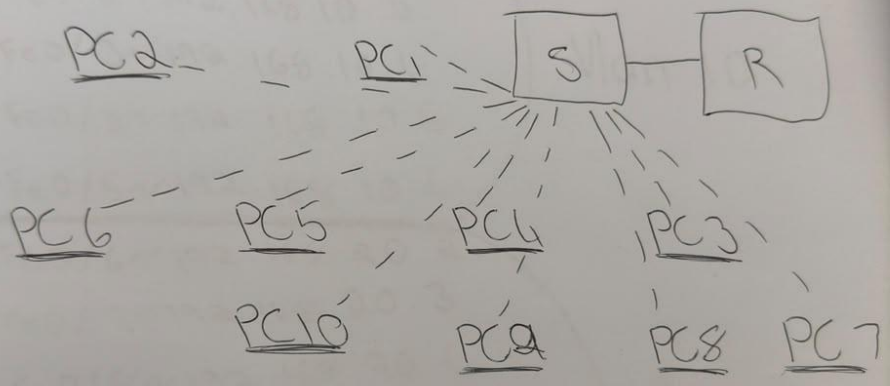
PC1 = Fe0/1 ~ 192.168.10.2
PC2 = Fe0/2 ~ 192.168.10.3
PC3 = Fe0/3 ~ 192.168.10.4
PC4 = Fe0/6 ~ 192.168.10.5
PC5 = Fe0/5 ~ 192.168.10.6

Vlan 10

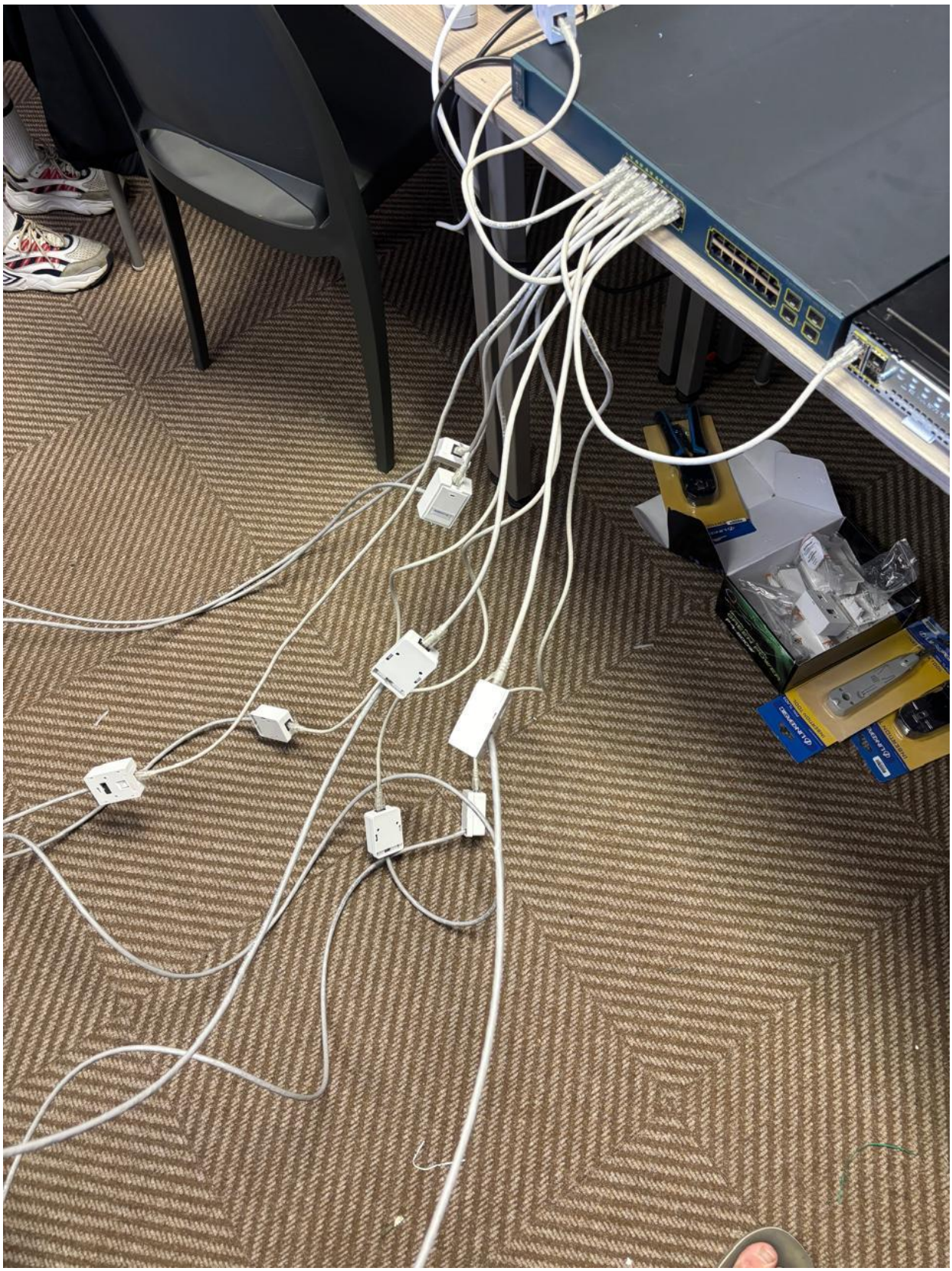
PC6 = Fe0/6 ~ 192.168.20.2
PC7 = Fe0/7 ~ 192.168.20.3
PC 8 = Fe0/8 ~ 192.168.20.4
PC 9 = Fe0/9 ~ 192.168.20.5
PC 10 = Fe0/10 ~ 192.168.20.6

Vlan 20

Topology







1. Install switch cabinet.
2. Install patch panels.
3. Configure switches.
4. Configure routers.
5. Implement VLANs.
6. Configure printers.
7. Test and troubleshoot the network.

Introduced Faults

- Incorrect VLAN assignment
- Incorrect subnet mask
- Incorrect default gateway
- Duplicate IP address
- Faulty patch lead
- Disabled switch port
- Router interface shutdown
- Missing route

Troubleshooting Commands

```
show running-config
show vlan brief
show interfaces
show ip route
show ip interface brief
show mac address-table
ping
tracert
```

Deliverable

Completed troubleshooting report.

Assessment Rubric (10 Marks)

Criteria	Marks
Fault identification	3
Troubleshooting methodology	3
Correct resolution	2
Professional report	2

Total	10

PRACTICAL OBSERVATION CHECKLIST

Competency	Yes	No
Uses tools safely	✔	☐
Uses crimping tools correctly	✔	☐
Uses punchdown tools correctly	✔	☐
Applies T568B correctly	✔	☐
Configures Cisco devices correctly	✔	☐
Demonstrates troubleshooting skills	✔	☐
Maintains clean workstation	✔	☐
Demonstrates teamwork	✔	☐
Produces accurate documentation	✔	☐